

## 43.3 Penalized splines

The last section ended with a dilemma. A regression spline with few knots is too rigid to follow the data; one with many knots overfits; and the number of knots is a discrete parameter that cannot be tuned gently. The resolution is to stop treating the knots as the control: take *too many* of them — enough that the basis could interpolate — and shrink the coefficients towards a smooth configuration with a quadratic penalty. This is Definition 30.1.1 with a particular penalty, and by Theorem 27.2.2 it is ridge regression in the B-spline basis.

### 43.3.1 The difference penalty

If the basis is rich, neighbouring B-splines overlap heavily and neighbouring coefficients describe nearly the same part of the curve. A wiggly fit is one whose coefficient sequence  $\gamma_1, \dots, \gamma_m$  jumps about, a smooth fit one whose coefficients change slowly with  $j$ , so penalizing the differences of the coefficients penalizes roughness — with no integrals to evaluate, since the penalty is a quadratic form whose matrix is known in advance.

Write  $\Delta\gamma_j = \gamma_j - \gamma_{j-1}$  and  $\Delta^k = \Delta(\Delta^{k-1})$ , so that  $\Delta^2\gamma_j = \gamma_j - 2\gamma_{j-1} + \gamma_{j-2}$ . Let  $\mathbf{D}_k$  be the  $(m-k) \times m$  matrix with  $(\mathbf{D}_k\boldsymbol{\gamma})_j = \Delta^k\gamma_{j+k}$ ; for  $k=1$  and  $k=2$  its rows are  $(\dots, -1, 1, \dots)$  and  $(\dots, 1, -2, 1, \dots)$ .

#### Definition 43.3.1 — Penalized spline

Let  $\mathbf{B}$  be the B-spline basis matrix of Definition 43.2.1 for  $K$  interior knots, equally spaced, with  $K$  chosen generously (say 20 to 40, and in any case no more than  $n$ ). For  $\lambda \geq 0$  and an integer  $k \geq 1$ , the **penalized spline** or **P-spline** estimate is the function  $\hat{f} = \sum_j \hat{\gamma}_j B_j^d$  whose coefficients minimize

$$\|\mathbf{y} - \mathbf{B}\boldsymbol{\gamma}\|^2 + \lambda \|\mathbf{D}_k\boldsymbol{\gamma}\|^2 = \|\mathbf{y} - \mathbf{B}\boldsymbol{\gamma}\|^2 + \lambda \boldsymbol{\gamma}^\top \mathbf{P}\boldsymbol{\gamma}, \quad \mathbf{P} = \mathbf{D}_k^\top \mathbf{D}_k. \quad (43.3.1)$$

$\lambda$  is the **smoothing parameter** and  $k$  the **order** of the penalty.

This is the construction of Eilers and Marx (1996), who gave it the name P-spline. In the scaling of Definition 30.1.1 the criterion carries a factor  $\frac{1}{2}$  and  $P(\boldsymbol{\gamma}) = \frac{1}{2}\boldsymbol{\gamma}^\top \mathbf{P}\boldsymbol{\gamma}$ ; nothing depends on the convention except the numerical value of  $\lambda$ .

Three things have changed since Section 43.2. The number of knots is a computational decision, not a modelling one: take enough. The estimate is no longer a projection, so it has bias even when  $f$  lies in the spline space. And the dial is continuous.

### 43.3.2 The fit, and what the penalty does to it

#### Lemma 43.3.1 — The two matrices diagonalize together

Suppose  $\mathbf{B}$  has full column rank  $m$ , and let  $\mathbf{P}$  be nonnegative definite. There is a nonsingular  $\mathbf{T}$  with  $\mathbf{T}^\top \mathbf{B}^\top \mathbf{B} \mathbf{T} = \mathbf{I}$  and  $\mathbf{T}^\top \mathbf{P} \mathbf{T} = \text{diag}(e_1, \dots, e_m)$ , where  $0 \leq e_1 \leq$

$\dots \leq e_m$  and exactly  $\text{rank}(\mathbf{P})$  of the  $e_j$  are positive. The columns  $\mathbf{u}_1, \dots, \mathbf{u}_m$  of  $\mathbf{BT}$  are orthonormal.

*Proof.*  $\mathbf{B}^\top \mathbf{B}$  is positive definite, so Corollary 1.7.8(b) supplies a nonsingular  $\mathbf{T}$  with  $\mathbf{T}^\top \mathbf{B}^\top \mathbf{B} \mathbf{T} = \mathbf{I}$  and  $\mathbf{T}^\top \mathbf{P} \mathbf{T}$  diagonal. Its entries are nonnegative because  $\mathbf{P}$  is nonnegative definite, and the number of nonzero ones is  $\text{rank}(\mathbf{T}^\top \mathbf{P} \mathbf{T}) = \text{rank}(\mathbf{P})$ ; permuting the columns of  $\mathbf{T}$  puts them in increasing order and preserves both identities. Orthonormality of the columns of  $\mathbf{BT}$  is the statement  $(\mathbf{BT})^\top (\mathbf{BT}) = \mathbf{I}$ . ■

The vectors  $\mathbf{u}_j$  are the **Demmler–Reinsch basis** of the smoother. Ordered by  $e_j$ , they run from smooth to rough: those with  $e_j = 0$  are the shapes the penalty does not see at all.

### Proposition 43.3.2 — A P-spline is a ridge regression

Let  $\mathbf{B}$  have full column rank and  $\lambda > 0$ , and write  $\mathbf{T}, e_j, \mathbf{u}_j$  as in Lemma 43.3.1.

- (a)  $\hat{\mathbf{y}} = (\mathbf{B}^\top \mathbf{B} + \lambda \mathbf{P})^{-1} \mathbf{B}^\top \mathbf{y}$  is the unique minimizer of Eq. 43.3.1, and the fitted vector is  $\hat{\mathbf{f}} = \mathbf{S}_\lambda \mathbf{y}$  with

$$\mathbf{S}_\lambda = \mathbf{B}(\mathbf{B}^\top \mathbf{B} + \lambda \mathbf{P})^{-1} \mathbf{B}^\top = \sum_{j=1}^m \frac{1}{1 + \lambda e_j} \mathbf{u}_j \mathbf{u}_j^\top. \quad (43.3.2)$$

- (b)  $\mathbf{S}_\lambda$  is symmetric with eigenvalues  $(1 + \lambda e_j)^{-1} \in (0, 1]$  and the same eigenvectors for every  $\lambda$ . Its **effective degrees of freedom** — the count of Theorem 27.2.2(b), now for a roughness penalty —

$$\text{df}(\lambda) = \text{tr}(\mathbf{S}_\lambda) = \sum_{j=1}^m \frac{1}{1 + \lambda e_j} \quad (43.3.3)$$

decrease strictly and continuously from  $m$  at  $\lambda = 0$  to  $m - \text{rank}(\mathbf{P})$  as  $\lambda \rightarrow \infty$ . For a  $k$ th-order difference penalty,  $\text{rank}(\mathbf{P}) = m - k$ , so the limit is  $k$ .

- (c) As  $\lambda \rightarrow \infty$ ,  $\mathbf{S}_\lambda$  tends to the orthogonal projection onto  $\mathcal{C}\{\mathbf{B}\mathcal{N}(\mathbf{P})\}$ , a space of dimension  $k$ .

*Proof.*

- (a) The criterion is  $\boldsymbol{\gamma}^\top (\mathbf{B}^\top \mathbf{B} + \lambda \mathbf{P}) \boldsymbol{\gamma} - 2\boldsymbol{\gamma}^\top \mathbf{B}^\top \mathbf{y} + \mathbf{y}^\top \mathbf{y}$ , and  $\mathbf{B}^\top \mathbf{B} + \lambda \mathbf{P}$  is positive definite because  $\mathbf{B}^\top \mathbf{B}$  is and  $\mathbf{P}$  is nonnegative definite; Proposition 1.11.2 gives the unique minimizer. In the coordinates  $\boldsymbol{\gamma} = \mathbf{T}\boldsymbol{\alpha}$  the criterion becomes  $\|\mathbf{y} - \mathbf{B}\mathbf{T}\boldsymbol{\alpha}\|^2 + \lambda \sum_j e_j \alpha_j^2$ , and because the columns of  $\mathbf{BT}$  are orthonormal this separates:  $\|\mathbf{y}\|^2 - \sum_j \{2\alpha_j \mathbf{u}_j^\top \mathbf{y} - \alpha_j^2 (1 + \lambda e_j)\}$  plus the part of  $\mathbf{y}$  orthogonal to all the  $\mathbf{u}_j$ . Minimizing term by term gives  $\hat{\alpha}_j = \mathbf{u}_j^\top \mathbf{y} / (1 + \lambda e_j)$  and  $\hat{\mathbf{f}} = \sum_j \hat{\alpha}_j \mathbf{u}_j$ , which is Eq. 43.3.2.
- (b) The representation Eq. 43.3.2 is a spectral decomposition, since the  $\mathbf{u}_j$  are orthonormal; the trace is the sum of the eigenvalues. Each term  $(1 + \lambda e_j)^{-1}$  is constant in  $\lambda$  if  $e_j = 0$  and strictly decreasing to zero otherwise, and there are  $\text{rank}(\mathbf{P})$  of the latter. For the difference penalty,  $\mathbf{D}_k$  has  $m - k$  rows and full row rank (its leading  $(m - k) \times (m - k)$  submatrix is triangular with  $\pm 1$  on the diagonal), so  $\text{rank}(\mathbf{P}) = \text{rank}(\mathbf{D}_k) = m - k$ .

- (c) By Eq. 43.3.2,  $\mathbf{S}_\lambda \rightarrow \sum_{j:e_j=0} \mathbf{u}_j \mathbf{u}_j^\top$ , the orthogonal projection onto the span of those  $\mathbf{u}_j$ . Those columns of  $\mathbf{B}\mathbf{T}$  are  $\mathbf{B}$  times the columns of  $\mathbf{T}$  with  $\mathbf{T}^\top \mathbf{P}\mathbf{T}$ -entry zero, and  $\mathbf{T}^\top \mathbf{P}\mathbf{T}\mathbf{e}_j = 0$  with  $\mathbf{T}$  nonsingular means  $\mathbf{T}\mathbf{e}_j \in \mathcal{N}(\mathbf{P})$ ; counting dimensions gives all of  $\mathcal{N}(\mathbf{P})$ . ■

*Idea.* Compare Theorem 27.2.2(b). Ridge shrinks the coordinate along the  $j$ th singular direction by  $d_j^2/(d_j^2 + \lambda)$ , and the small singular values are shrunk most. A P-spline shrinks the coordinate along  $\mathbf{u}_j$  by  $1/(1 + \lambda e_j)$ , and the *rough* directions are shrunk most. The difference is only in what the geometry calls “small”: ridge measures size by the design, a smoother measures it by roughness. Both count what is left with a trace.

### 43.3.3 What survives as the penalty grows

Part (c) says the limit is a projection onto a  $k$ -dimensional space. Which one?

#### Proposition 43.3.3 — The limiting fit

Let  $\mathbf{D}_k$  be the  $k$ th difference matrix.

- (a)  $\mathcal{N}(\mathbf{D}_k) = \{\boldsymbol{\gamma} : \gamma_j = p(j) \text{ for a polynomial } p \text{ of degree } < k\}$ , of dimension  $k$ .
- (b) For  $k = 1$ , the limiting fit of Proposition 43.3.2(c) is the constant  $\bar{y}$ , whatever the knots.
- (c) For  $k = 2$  and equally spaced Greville abscissae (Proposition 43.2.2) — the uniform knot sequence of Section 43.2 — the limiting fit is the ordinary least squares line.

*Proof.*

- (a) If  $\gamma_j = p(j)$  with  $\deg p < k$  then  $\Delta$  lowers the degree by one, so  $\Delta^k \boldsymbol{\gamma} = \mathbf{0}$ . Conversely the set of such  $\boldsymbol{\gamma}$  has dimension  $k$  (the polynomials  $1, j, \dots, j^{k-1}$  restricted to  $j = 1, \dots, m$  are independent as long as  $m \geq k$ ), and  $\dim \mathcal{N}(\mathbf{D}_k) = m - \text{rank}(\mathbf{D}_k) = k$  by the proof of Proposition 43.3.2(b). Two subspaces of the same dimension, one inside the other, coincide.
- (b) For  $k = 1$ ,  $\mathcal{N}(\mathbf{D}_1)$  is spanned by  $\mathbf{1}$ , and  $\mathbf{B}\mathbf{1}$  is the vector of values of  $\sum_j B_j^d$ , which is  $\mathbf{1}$  by Proposition 43.2.1(c). The projection onto  $\text{span}\{\mathbf{1}\}$  sends  $\mathbf{y}$  to  $\bar{y}\mathbf{1}$ .
- (c) For  $k = 2$ ,  $\mathcal{N}(\mathbf{D}_2)$  is spanned by  $\mathbf{1}$  and  $\mathbf{j} = (1, \dots, m)^\top$ . If the Greville abscissae satisfy  $\xi_j = \xi_1 + (j-1)\delta$ , then  $\mathbf{j} = \{\mathbf{1}(\delta - \xi_1) + \boldsymbol{\xi}\}/\delta$  with  $\boldsymbol{\xi} = (\xi_1, \dots, \xi_m)^\top$ , so  $\text{span}\{\mathbf{1}, \mathbf{j}\} = \text{span}\{\mathbf{1}, \boldsymbol{\xi}\}$ . By Proposition 43.2.1(c) and Eq. 43.2.5,  $\mathbf{B}\mathbf{1}$  is the vector of ones and  $\mathbf{B}\boldsymbol{\xi}$  is the vector  $(x_1, \dots, x_n)^\top$ . So  $\mathcal{C}\{\mathbf{B}\mathcal{N}(\mathbf{P})\}$  is the column space of  $[\mathbf{1}, \mathbf{x}]$ , and the projection onto it is the least squares line. ■

*Remark* (Why the knot sequence matters here and nowhere else). Part (c) is exactly true for the uniform sequence and only approximately true for the clamped sequence Eq. 43.2.1, whose Greville abscissae bunch up near the endpoints. On the Nile data of

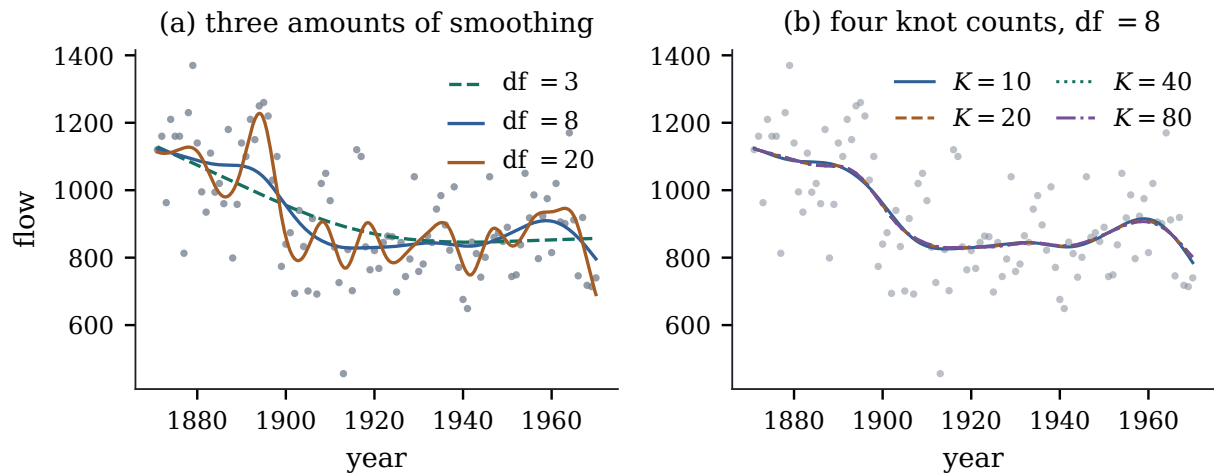


Figure 43.3.1: Annual Nile flow. (a) Cubic P-splines with a second-order penalty, labelled by effective degrees of freedom Eq. 43.3.3. (b) The same fit at  $df = 8$  with  $K = 10, 20, 40$  and  $80$  interior knots: four curves are drawn and one is visible.

**Example (Nile flow, 1871 to 1970)**, against a response range of 914, the uniform fit at  $\lambda = 10^{12}$  departs from the least squares line by  $3.2 \times 10^{-2}$  — the numerical remainder of a penalty that is large but finite — while the clamped fit departs by 11.4, some 1.2 per cent of that range: visually almost nothing, but not the line. Since the two sequences span the same functions, the choice affects only what “the penalty is zero” means. For  $k \geq 3$  the limiting space is a  $k$ -dimensional space of splines containing the straight lines, and consists exactly of the polynomials of degree  $k - 1$  when the knots are equally spaced (Eilers and Marx, 1996; de Boor, 2001, ch. IX), a fact we use but do not prove.

### 43.3.4 Knots stop mattering

#### Example. (Nile flow, 1871 to 1970)

The annual flow of the Nile at Aswan was recorded for 100 years from 1871 to 1970. Rescale the year to  $[0, 1]$  and fit cubic P-splines with a second-order penalty. Reaching 3, 8 and 20 effective degrees of freedom needs  $\lambda = 9.1 \times 10^2$ ,  $4.7 \times 10^0$  and  $1.5 \times 10^{-2}$ : five orders of magnitude for a threefold change in  $df$ , which is why  $\lambda$  is searched on a logarithmic scale and reported as  $df$ .

Now hold  $df$  at 8 and change the number of knots. With  $K = 10, 20, 40$  and  $80$  — the last giving 84 coefficients for 100 observations — the four curves differ nowhere by more than 16.60, or 1.82 per cent of the response range, and the  $K = 40$  and  $K = 80$  curves by at most 1.373. Without the penalty the same knot counts give curves differing by up to 258.2, fifteen times as much.

Figure 43.3.1 draws both comparisons. The fitted curve descends steeply around 1900, and it is worth saying what a smoother does and does not know: the historical record is of an abrupt change, and a smoother with a second-order penalty cannot represent a jump. It renders one as a steep smooth descent and reports no lack of fit (Theorem 14.6.2 would, if the design had replicates). Smoothness is an assumption, and like every assumption in this book it buys precision by excluding something.

```

import numpy as np
import statsmodels.api as sm

def uniform_knots(a, b, K, d):
    """K interior knots equally spaced in (a, b), continued d further steps at each
    end."""
    delta = (b - a) / (K + 1)
    return a + delta * np.arange(-d, K + 2 + d)

def bspline_basis(x, kn, d):
    """The B-splines of degree d on the knot vector kn, by the de Boor recurrence.
    """
    x = np.atleast_1d(np.asarray(x, float))
    B = np.array([(x >= kn[j]) & (x < kn[j + 1]) for j in range(len(kn) - 1)], float)
    B.T
    last = np.max(np.nonzero(kn[:-1] < kn[1:])[0]) # close the rightmost
    interval
    B[x == kn[-1], last] = 1.0
    for deg in range(1, d + 1):
        C = np.zeros((len(x), B.shape[1] - 1))
        for j in range(C.shape[1]):
            if kn[j + deg] > kn[j]:
                C[:, j] += (x - kn[j]) / (kn[j + deg] - kn[j]) * B[:, j]
            if kn[j + deg + 1] > kn[j + 1]:
                C[:, j] += (kn[j + deg + 1] - x) / (kn[j + deg + 1] - kn[j + 1]) * B
           [:, j + 1]
        B = C
    return B

def difference_matrix(m, k):
    """The k-th difference operator on m coefficients, an (m-k) x m matrix."""
    return np.diff(np.eye(m), n=k, axis=0)

def pspline(x, y, lam, K=20, d=3, k=2, a=None, b=None):
    """Penalized spline fit: minimize ||y - B gamma||^2 + lam ||Delta^k gamma||^2.
    """
    a = x.min() if a is None else a
    b = x.max() if b is None else b
    kn = uniform_knots(a, b, K, d)
    B = bspline_basis(x, kn, d)
    Dk = difference_matrix(B.shape[1], k)
    A = B.T @ B + lam * Dk.T @ Dk
    gamma = np.linalg.solve(A, B.T @ y)
    df = np.trace(np.linalg.solve(A, B.T @ B)) # trace of the smoother
    matrix
    return gamma, df, kn, d

def evaluate(gamma, kn, d, grid):
    return bspline_basis(grid, kn, d) @ gamma

data = sm.datasets.nile.load_pandas().data

```

```

year = data["year"].to_numpy()
y = data["volume"].to_numpy()
n = len(y)
x = (year - year.min()) / (year.max() - year.min())      # the covariate, rescaled
    to [0, 1]
grid = np.linspace(0, 1, 401)

gamma_big, df_big, kn, d = pspline(x, y, 1e12)
line = np.polyval(np.polyfit(x, y, 1), grid)
print("df at lambda = 1e12:", round(df_big, 6))
print("largest gap from the least squares line:", np.max(np.abs(evaluate(gamma_big,
    kn, d, grid) - line)))

```

[Run this code online](#)

### 43.3.5 Choices that remain

**How many knots.** Enough that adding more changes nothing, which [Example \(Nile flow, 1871 to 1970\)](#) suggests is a few dozen; Ruppert's rule of thumb is  $K = \min(n/4, 35)$ , and by Exercise (B2) generosity is cheap.

**The degree.** Cubic ( $d = 3$ ) throughout; higher degrees buy nothing once a penalty is present.

**The order of the penalty.**  $k = 2$  is the default, because its null space is the straight lines: when the data say there is no curvature, the fit degrades to the linear model of Part II rather than to a horizontal line ( $k = 1$ ) or a parabola ( $k = 3$ ). The order also fixes the behaviour *beyond* the data, where the penalty is all there is: a  $k$ th-order penalty extrapolates as a polynomial of degree  $k - 1$ , one more reason to distrust an extrapolated smooth (Proposition 42.1.1).

**The smoothing parameter.** [Section 43.5](#).

*Remark* (Derivative penalties). A different penalty measures roughness by an integral,  $\int_a^b \{f''(t)\}^2 dt = \boldsymbol{\gamma}^\top \boldsymbol{\Omega} \boldsymbol{\gamma}$  with  $\Omega_{jk} = \int B_j'' B_k''$ , computable exactly from Eq. 43.2.3 and Gauss–Legendre quadrature and banded for the same reason  $\mathbf{B}^\top \mathbf{B}$  is. This is the O'Sullivan (1986) penalty, and it makes the P-spline a finite-dimensional version of the smoothing spline of the next section; Wand and Ormerod (2008) show that for equally spaced knots [Definition 43.3.1](#)'s penalty is a difference approximation to it, which is why the two fits look identical. The difference penalty is simpler to state and easier to generalize — nothing in [Eq. 43.3.1](#) mentions an integral — and is the one used in [Chapter 44](#).

### 43.3.6 Exercises

#### 43.3.6.1 A. Check your understanding

##### Exercise A1.

Using [Eq. 43.3.3](#), sketch  $df(\lambda)$  against  $\log \lambda$  for  $m = 24$  and a second-order penalty. What are the two horizontal asymptotes, and why is  $\log \lambda$  the right scale for the horizontal axis?

## 43.3.6.2 B. Practice

**Exercise B1.**

A model with 8 parameters uses 8 degrees of freedom; a P-spline at  $\text{df}(\lambda) = 8$  uses 24 parameters. Explain, using [Eq. 43.3.2](#), in what sense the two are comparable, and show that  $\text{tr}(\mathbf{S}_\lambda \mathbf{S}_\lambda^\top) \leq \text{tr}(\mathbf{S}_\lambda)$  with equality only when  $\mathbf{S}_\lambda$  is a projection. (Both traces are used as degrees of freedom in the literature; see [Section 43.5](#).)

*Solution.* By [Eq. 43.3.2](#),  $\mathbf{S}_\lambda$  has eigenvalues  $s_j = (1 + \lambda e_j)^{-1} \in (0, 1]$ , so  $\text{tr}(\mathbf{S}_\lambda) = \sum_j s_j$  and  $\text{tr}(\mathbf{S}_\lambda \mathbf{S}_\lambda^\top) = \sum_j s_j^2 \leq \sum_j s_j$ , with equality iff every  $s_j \in \{0, 1\}$ , that is iff  $\mathbf{S}_\lambda$  is a projection (Theorem 6.3.3). A least squares fit on  $p$  columns has  $p$  eigenvalues equal to one; a smoother spreads the same total over more directions, using each only partly. This is the covariance-penalty degrees of freedom Definition 29.1.2 applied to a linear rule.

**Exercise B2.**

Show that the P-spline criterion [Eq. 43.3.1](#) is Theorem 29.3.2's criterion  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \mathbf{b}^\top \mathbf{\Omega} \mathbf{b}$  with  $\mathbf{X} = \mathbf{B}$  and  $\mathbf{\Omega} = \lambda \mathbf{P}$ , and conclude that the leave-one-out shortcut  $(y_i - \hat{f}(x_i))/(1 - s_{ii})$  is exact for a P-spline. What changes if the knots are chosen by looking at the data first?

## 43.3.6.3 C. Going deeper

**Exercise C1.**

Compute the Demmler–Reinsch basis of [Lemma 43.3.1](#) for a cubic P-spline with  $K = 20$ ,  $k = 2$ , on 100 equally spaced design points, and plot  $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_5, \mathbf{u}_{10}$  against  $x$ . Confirm that  $e_1 = e_2 = 0$  and that the corresponding  $\mathbf{u}_j$  span the constant and linear functions, and that  $\mathbf{u}_j$  crosses zero more often as  $e_j$  grows. Relate the picture to [Eq. 43.3.2](#): which shapes survive a large  $\lambda$ ?

**Exercise C2.**

[Proposition 43.3.2](#) assumes  $\mathbf{B}$  has full column rank, which fails if some knot interval contains no design point. Show that the minimizer of [Eq. 43.3.1](#) is still unique provided  $\mathcal{N}(\mathbf{B}) \cap \mathcal{N}(\mathbf{P}) = \{\mathbf{0}\}$ , and give an example with  $K$  large and a gap in the design where this fails for  $k = 1$  but holds for  $k = 2 \dots$  or show that it cannot fail for  $k = 2$  either. (Hint: Theorem 30.1.1(c), and the fact that  $\mathcal{N}(\mathbf{P}) = \mathcal{N}(\mathbf{D}_k)$ .)

*Solution.*  $\mathbf{B}^\top \mathbf{B} + \lambda \mathbf{P}$  is nonnegative definite, and  $\boldsymbol{\gamma}^\top (\mathbf{B}^\top \mathbf{B} + \lambda \mathbf{P}) \boldsymbol{\gamma} = 0$  forces  $\mathbf{B}\boldsymbol{\gamma} = \mathbf{0}$  and  $\mathbf{D}_k \boldsymbol{\gamma} = \mathbf{0}$ ; so the matrix is positive definite, and the minimizer unique, exactly when the two null spaces meet only at zero. This is Theorem 30.1.1(c) with a strictly convex penalty on the relevant subspace. For  $k = 2$ ,  $\boldsymbol{\gamma} \in \mathcal{N}(\mathbf{D}_2)$  means  $\gamma_j = \alpha + \beta j$ , and by [Proposition 43.3.3\(c\)](#) the corresponding function is  $\alpha' + \beta' x$  with  $\beta' \propto \beta$ ; such a function vanishes at all  $n \geq 2$  distinct design points only if  $\alpha' = \beta' = 0$ . So the intersection is trivial for  $k = 2$  whatever the gaps, and the fit is always unique. For  $k = 1$  the same argument with constants gives the same conclusion. The moral is that a penalty repairs

rank deficiency exactly on its own null space — which is why a generous  $K$  is safe.

### Exercise C3.

Assume  $E(\mathbf{Y}) = \mathbf{f}$  and  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ . Using Eq. 43.3.2, show that

$$E\|\hat{\mathbf{f}} - \mathbf{f}\|^2 = \sum_j \left( \frac{\lambda e_j}{1 + \lambda e_j} \right)^2 (\mathbf{u}_j^\top \mathbf{f})^2 + \sigma^2 \sum_j \frac{1}{(1 + \lambda e_j)^2} + \|(\mathbf{I} - \mathbf{U}\mathbf{U}^\top)\mathbf{f}\|^2,$$

where  $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_m]$ . Identify the three terms, and say which of them no choice of  $\lambda$  can reduce.

*Solution.* Write  $\mathbf{f} = \mathbf{U}\mathbf{U}^\top \mathbf{f} + (\mathbf{I} - \mathbf{U}\mathbf{U}^\top)\mathbf{f}$ . Since  $\mathbf{S}_\lambda = \mathbf{U} \text{diag}(s_j) \mathbf{U}^\top$  with  $s_j = (1 + \lambda e_j)^{-1}$ ,  $\hat{\mathbf{f}} - \mathbf{f} = \mathbf{U}\{\text{diag}(s_j)\mathbf{U}^\top(\mathbf{f} + \boldsymbol{\varepsilon}) - \mathbf{U}^\top \mathbf{f}\} - (\mathbf{I} - \mathbf{U}\mathbf{U}^\top)\mathbf{f}$ , and the three pieces are orthogonal in expectation. The first sum is the squared shrinkage bias,  $(s_j - 1)\mathbf{u}_j^\top \mathbf{f} = -\lambda e_j \mathbf{u}_j^\top \mathbf{f} / (1 + \lambda e_j)$ ; the second the variance  $\sigma^2 \sum_j s_j^2$ ; the third the *approximation* error, the part of  $\mathbf{f}$  outside the spline space, which does not involve  $\lambda$ . Only a richer basis reduces it, which is the argument for a generous  $K$ .